

PROJECT PROPOSAL

CyberSafe Kids

An Edutainment Platform for Cybercrime Awareness in Children

Course	IST 606 Security Management and Computer Forensics
Submitted To	Dr Ali Waka
Date	May 2026
Project Phase	Phase 1 Children (Ages 6–12)
Status	Proposal Draft for Review

1. Project Topic

CyberSafe Kids: Leveraging Edutainment to Educate Children on Cybercrime Awareness, Prevention, and Digital Safety

This project proposes the design and development of an interactive edutainment platform that uses engaging, age-appropriate content, including games, animated stories, and interactive scenarios, to teach children between the ages of 6 and 12 about cyber threats, safe online behaviours, and how to respond to cybercrime.

The initiative is framed as the first phase of a broader vision to extend cybercrime education across multiple demographics, from school-age children to corporate organisations.

2. Problem Statement

2.1 Background

The rapid expansion of internet access among children has created an environment in which young users are increasingly exposed to cyber threats. Children today use smartphones, tablets, and computers for education, entertainment, and social interaction, often without sufficient awareness of the digital dangers they face.

Cybercriminals are known to specifically target children due to their perceived vulnerability, trust, and limited understanding of online risks. Despite this, most formal educational systems, particularly in developing countries, have not integrated structured cybercrime awareness into their curricula.

2.2 The Core Problem

Children represent one of the most digitally active yet least cybersecurity-educated segments of society. The key challenges include:

- Children lack foundational knowledge of what constitutes a cyber threat (e.g., phishing, cyberbullying, online grooming, malware).
- Existing cybersecurity materials are largely designed for adults and are inappropriate in language, format, and complexity for children.
- Traditional classroom instruction on digital safety is either absent or insufficiently engaging for young learners.
- Parents and guardians are often not equipped to provide adequate digital safety guidance to their children.
- Consequences of cybercrime exposure for children include identity theft, psychological harm from cyberbullying, exploitation, and financial loss to families.

2.3 Justification

Research consistently shows that early education on digital safety significantly reduces the likelihood of falling victim to cybercrime. Edutainment, the combination of education and entertainment, is proven to be among the most effective approaches for knowledge acquisition in children, as it increases engagement, retention, and motivation to learn.

This project directly addresses the gap between growing cyber risk exposure among children and the near absence of accessible, engaging, and locally relevant cybercrime education tailored to their developmental stage.

3. Background Research and Findings

3.1 Global Cybercrime Trends Affecting Children

Recent studies and reports highlight the severity of cybercrime targeting children:

- The Internet Watch Foundation (IWF) reported that online child sexual abuse material increased by over 50% between 2019 and 2023, reflecting a sharp rise in online child exploitation.
- Cyberbullying affects approximately 37% of young people aged 12–17 globally (Cyberbullying Research Centre, 2023), with significant impacts on mental health, school performance, and social development.
- A 2022 report by Kaspersky found that children are disproportionately affected by phishing attacks, malware downloads, and identity-related cybercrime, often through gaming platforms, social media, and messaging apps.
- UNICEF's State of the World's Children 2023 report emphasised that digital literacy, including cyber safety, must be treated as a fundamental right for children in the digital age.

3.2 Effectiveness of Edutainment in Education

- A study in the Journal of Educational Psychology (2021) found that game-based learning improves information retention by up to 40% in children aged 6–12 compared to traditional instruction.

- The Sesame Workshop demonstrated through decades of research that combining entertainment with educational content dramatically improves learning outcomes in young children.
- Interactive storytelling and role-playing scenarios have been identified by UNICEF and UNESCO as highly effective tools for teaching children about safety, both physical and digital.
- Platforms such as Google's 'Be Internet Awesome' (Interland) and the UK's Childnet International have demonstrated measurable success using games and interactive media to teach digital safety.

3.3 Review of Existing Solutions and Identified Gaps

Platform	Strengths	Gaps
Google Interland (Be Internet Awesome)	Gamified, visually engaging, covers key safety themes	English-only; no local/African context; limited parental tools
Childnet International (UK)	Rich resources, teacher guides, and age-specific content	Designed for UK schools; low relevance in African contexts
Local school curricula	Reaches large numbers of children through an established system	Cybercrime education largely absent or limited to textbook content, not interactive
NCSC / gov awareness campaigns	Backed by authority; wide reach via media	Primarily adult-focused; not child-friendly in format or language

The analysis above reveals a clear and actionable gap: there is no localised, interactive edutainment solution designed specifically to teach children in Sub-Saharan Africa about cybercrime. CyberSafe Kids is proposed to fill this gap.

4. Full Project Proposal

4.1 Project Overview

CyberSafe Kids is an interactive edutainment platform designed to educate children aged 6 to 12 about cybercrime and online safety. The platform will combine animated storytelling, mini-games, interactive quizzes, and scenario-based learning modules, delivered via a mobile and web application. All content will be contextualised for the local environment to maximise relatability and engagement.

4.2 Project Objectives

1. Design and develop an engaging, age-appropriate edutainment platform that teaches children foundational cybercrime awareness.
2. Cover key cybercrime topics relevant to children, including cyberbullying, phishing, online grooming, password safety, and safe social media use.

3. Evaluate the platform's effectiveness in improving cybercrime awareness and safe online behaviour among children through a structured pilot study.
4. Create a scalable framework that can be extended to other target groups — teenagers, parents, and organisations — in future development phases.

4.3 Scope of Phase 1

This proposal covers Phase 1, which focuses exclusively on children aged 6 to 12. The scope includes:

- Development of a minimum viable product (MVP) comprising at least five interactive learning modules.
- Topics covered: cyberbullying, phishing and scam awareness, safe passwords and account security, privacy and personal information protection, and safe internet navigation.
- Delivery via a cross-platform mobile application and web browser interface.
- A parent and guardian dashboard for monitoring learning progress.
- Pilot testing in at least two primary schools with pre- and post-assessments.

4.4 Target Audience

Audience	Role in This Phase
Primary users	Children aged 6–12 (primary school learners)
Secondary users	Parents and guardians who monitor their child's learning progress
Facilitators	Teachers and school administrators supporting the pilot programme
Future phases	Teenagers, university students, SMEs, and large organisations

4.5 Platform Modules

The platform will comprise five core interactive modules:

Module 1: CyberSafe World (Introduction)

An animated virtual world where children create a personalised avatar and navigate through different 'lands', each representing a cybercrime topic. Progress unlocks new areas, badges, and rewards, providing motivation to complete all modules.

Module 2: The Bully Blocker

An interactive branching storyline where children identify cyberbullying scenarios and learn the correct responses. Characters face real-world decisions and the game branches based on player choices, teaching empathy and resilience.

Module 3: The Phishing Fisher

A mini-game where children learn to identify fake emails, suspicious links, and online scams. Presented as a fishing game players 'catch' safe messages and 'throw back' dangerous ones. Builds threat recognition instincts in a playful format.

Module 4: Password Castle

Players build and defend a virtual castle by creating strong passwords. Weak passwords cause the castle walls to crumble; strong passwords reinforce them. Teaches best practices in password security through visual cause-and-effect feedback.

Module 5: Privacy Shield

A scenario-based game that teaches children what personal information should and should not be shared online, with whom, and under what circumstances. Addresses online grooming awareness in an age-appropriate manner.

4.6 Methodology

The project will follow an Agile development methodology structured in three major stages:

5. Research and Content Design Phase (Months 1–2)

- Conduct a literature review of cybercrime risks facing children and edutainment best practices.
- Hold stakeholder interviews with educators, child psychologists, and cybersecurity professionals.
- Develop content scripts, game designs, and defined learning objectives per module.

6. Development and Build Phase (Months 3–5)

- Complete UI/UX design using child-centred design principles (bright visuals, simple navigation, large touch targets).
- Develop the front-end (React Native for mobile, React.js for web) and back-end (Node.js, MongoDB) of the platform.
- Integrate gamification elements: badges, leaderboards, streaks, and a progress dashboard.
- Build the parent and guardian monitoring dashboard.

7. Pilot Testing and Evaluation Phase (Months 6–7)

- Deploy the platform in two primary schools with a target of 60–100 child participants.
- Administer pre- and post-assessments to measure knowledge gain and attitudinal change.
- Collect teacher and parent feedback through structured surveys.
- Analyse results, refine the platform, and prepare a final evaluation report.

4.7 Expected Outcomes

- A fully functional MVP of the CyberSafe Kids platform (web and mobile).
- Measurable improvement in cybercrime awareness among participating children (target: 60% or greater improvement on post-assessment scores compared to baseline).

- A validated, reusable content and design framework for extending the platform to other demographics.
- A pilot evaluation report with data-driven recommendations for full-scale deployment.
- A scalable technical architecture ready for Phase 2 development.

4.8 Proposed Project Timeline

Phase	Key Activities	Duration
1	Research, stakeholder interviews, content and module design	Months 1–2
2	UI/UX design, front-end and back-end platform development	Months 3–5
3	Pilot testing in schools, pre/post assessments, feedback collection	Months 6–7
4	Data analysis, platform refinement, final report and academic presentation	Month 8

4.9 Resources Required

- Development stack: React Native (mobile), React.js (web), Node.js (back-end), MongoDB (database).
- Design tools: Figma for UI/UX prototyping; animation software for character and scene design.
- Human resources: Project lead, 1–2 developers, UI/UX designer, cybersecurity content advisor, child education consultant.
- Institutional support: Agreements from at least two primary schools for pilot deployment.
- Devices: Development laptops and test devices (smartphones and tablets of varying screen sizes).

4.10 Risk Assessment and Mitigation

Risk	Likelihood	Mitigation Strategy
Low internet connectivity in pilot schools	Medium	Design offline-first functionality; cache modules locally on device
Difficulty securing school partnerships	Low	Engage school principals early; leverage the lecturer's academic and professional network
Content complexity mismatch for target age group	Medium	Involve a child education specialist in content design; conduct usability testing with children early in development
Scope creep beyond Phase 1 boundaries	Low	Maintain a clearly defined Phase 1 backlog; defer features for future

		phases through formal change control
--	--	--------------------------------------

4.11 Future Development Phases

While this proposal covers Phase 1 targeting children aged 6–12, the broader project vision encompasses:

- Phase 2: Teenagers (13–17) addressing social media risks, sexting, online radicalisation, and identity theft.
- Phase 3: Parents and guardians have tools to monitor and actively guide children's digital behaviour.
- Phase 4: Small and medium enterprises (SMEs) edutainment-based cybersecurity awareness for employees.
- Phase 5: Large organisations and corporations implement immersive cybercrime awareness and compliance training modules.

5. Conclusion

CyberSafe Kids addresses a critical and growing need: as children become ever more active participants in the digital world, their exposure to cybercrime increases while their awareness and preparedness remain dangerously low. Conventional educational approaches have consistently failed to engage children sufficiently on this topic.

By combining entertainment with structured cybersecurity education, this project offers a scalable, evidence-based, and locally relevant solution that is both effective and enjoyable for young learners. The platform is designed with longevity and expandability in mind, establishing a strong foundation for a comprehensive cybercrime awareness ecosystem that will ultimately reach individuals, families, and organisations.

This proposal requests approval to proceed to the research and content design phase, with full platform development and a pilot study to follow as outlined in the proposed timeline above.